



BK-MInD: Architecting an AI-Powered Lecture Learning System

Phase 2 Engineering Specifications,
Subsystem Design, and Cloud Deployment



Reporting Period:
Jan-Mar 2026



Framework:
Capstone Thesis
Phase 2



Workstreams:
Architecture (NMK)
| Infrastructure
(PDZ) |
Subsystems (NNK)

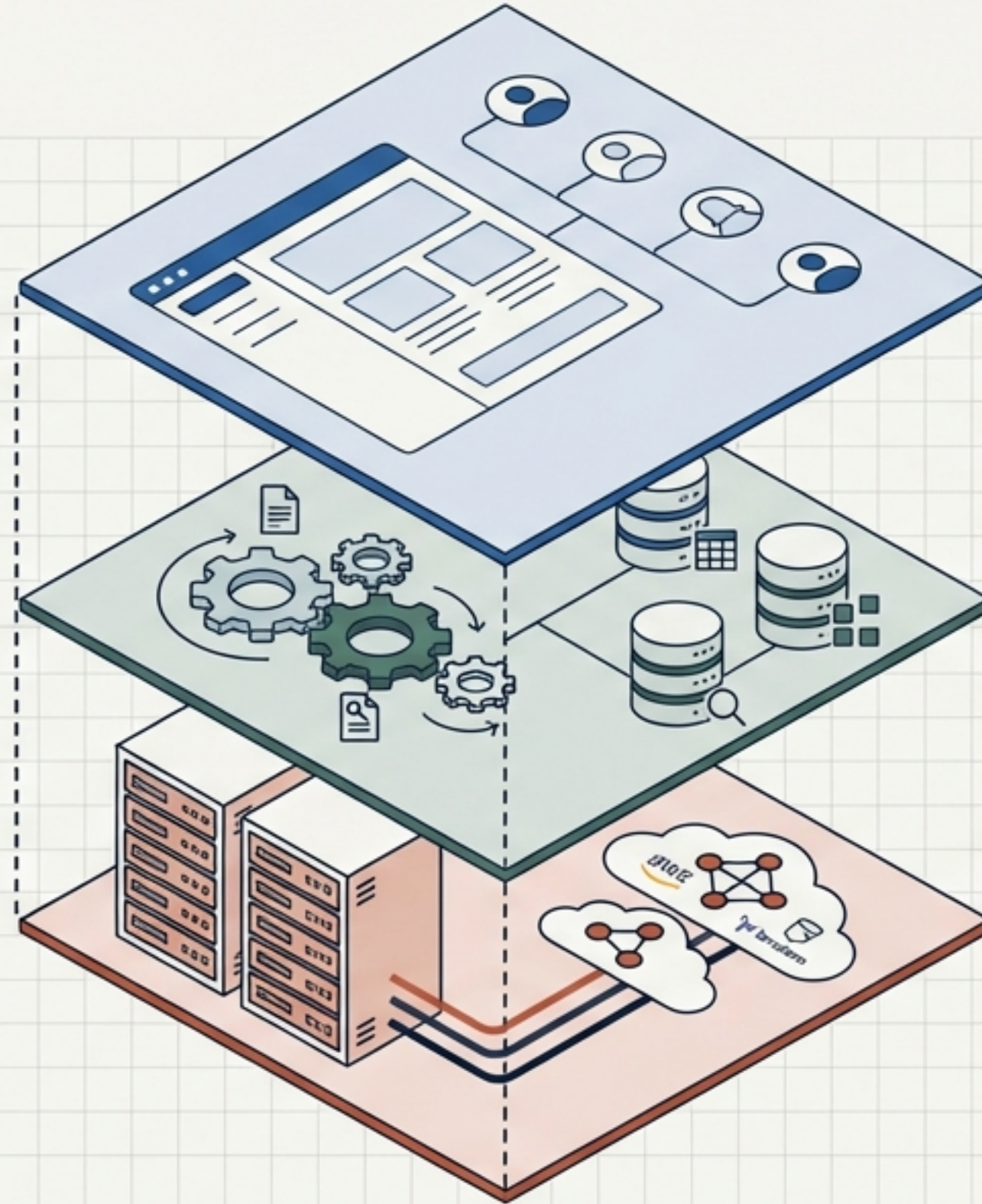
The System-Centric Architecture: Top-Down Translation

Layer 1: User Experience (Requirements & Use Cases)

Translating stakeholder needs into 35 rigorous engineering constraints and 10 interaction flows.

Layer 3: Hardware & Cloud Infrastructure (AWS & Terraform)

Cementing the system in reality with fully automated CI/CD and Terraform-provisioned AWS environments.



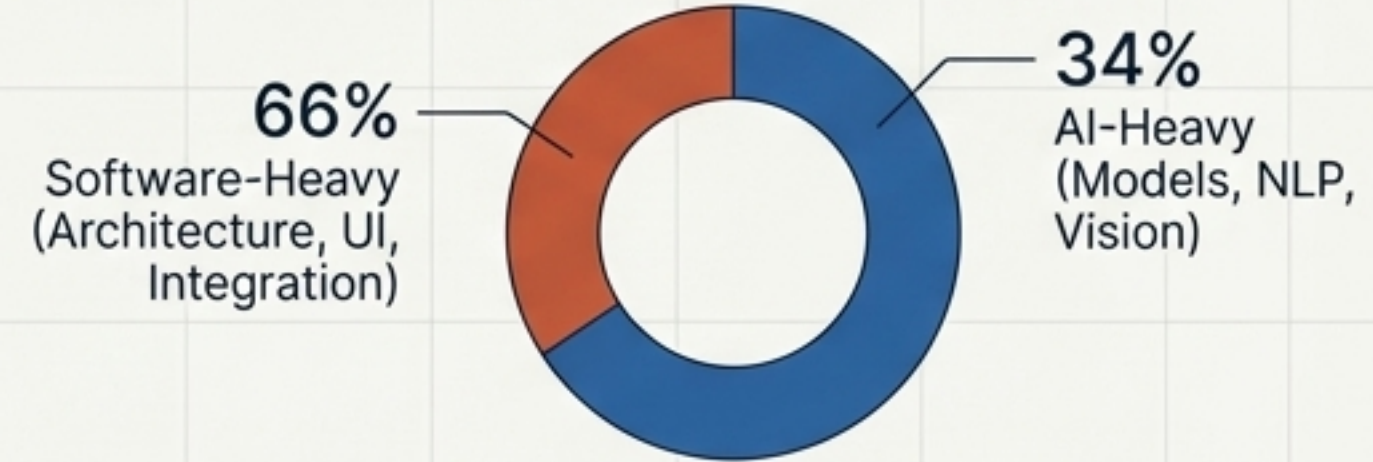
Layer 2: Application & AI Data Layer (Media Pipeline & Excel Parsing)

Converting unstructured multi-modal lectures and spreadsheets into highly structured, LLM-ready vector spaces.

Phase 2 transforms theoretical RAG prototypes into a fully documented, architecturally grounded, and operational cloud system.

Requirements Architecture Dashboard

System Balance



Tier 1: Functional (20)

- Content Processing (40%)
- Information Retrieval
- Question Answering
- User Interface
- Lecture Summaries
- Personalization

Tier 2: Non-Functional (8)

- Sub-second text retrieval
- 2-second visual queries
- 99% business-hour uptime
- 10,000+ document scale
- Graceful error recovery

Tier 3: Technical (7)

- Python 3.9+ / FastAPI
- React 18+ / Vite
- Vector & Document Stores
- Docker Compose Deployments
- Docker Compose Deployments
- Cloud Support (AWS/GCP/Azure)

Use Case Capability Matrix

Card 1: Core RAG System (Foundation)

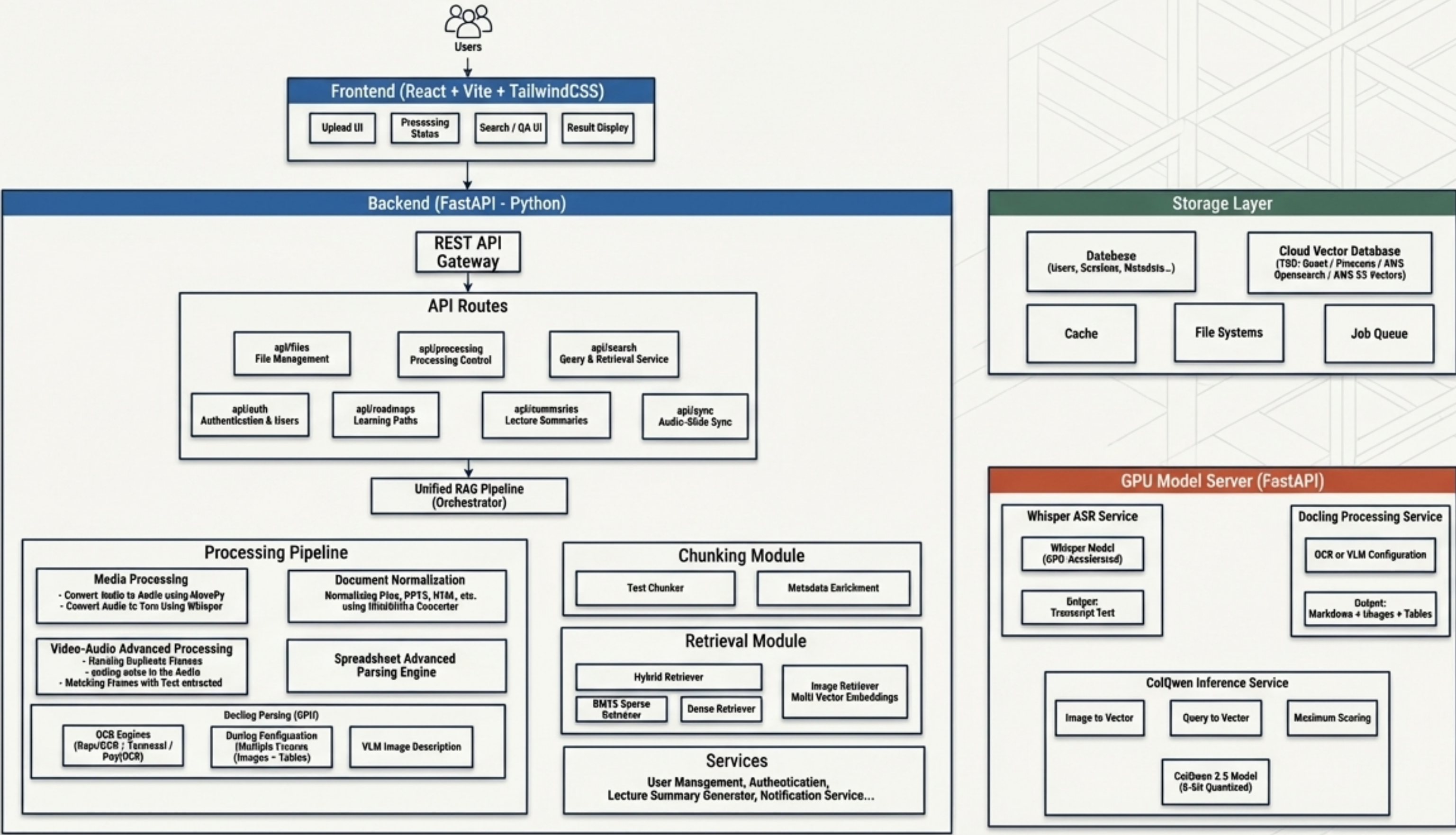
- UC-001: Upload & Process Content (Video, Audio, Docs)
- UC-002: Hybrid Search (Multi-modal retrieval)
- UC-003: Generate Answer with Citations (LLM context assembly)
- UC-004: Pipeline Management (Admin monitoring)

Parallel implementation
unblocked once foundation
is established.

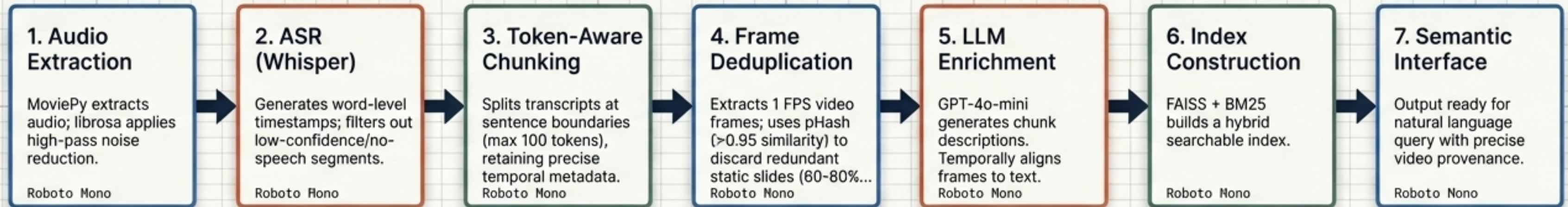
Card 2: Extended Learning Features (Evolution)

- UC-005/006: Automated Summaries & Interactive Navigation (Timestamp linking)
- UC-007: Custom Summaries (Difficulty/Topic focused)
- UC-008: Personalized Learning Paths (Knowledge gap analysis)
- UC-009/010: Adaptive Assessment & Analytics Dashboard

Logical System Architecture



7-Stage Media Processing Pipeline



Bridging the Semantic Gap in Media Ingestion

The Problem: Raw Transcript

So this thing over here,
this algorithm we talked
about last week...



Zero-similarity score for
student queries.

GPT-4o-mini
Processing
Node

The Solution: LLM-Enriched Description

This segment explains the
likelihood calculation in
Naive Bayes and how prior
probabilities are combined
with evidence.



Dramatic improvement in
retrieval relevance.

Hybrid Retrieval Strategy Comparison

Sparse Retrieval (BM25)

- **Mechanism:** Term-frequency/Inverse-document-frequency.
- **Strength:** Exact-match and strict keyword focus.
- **Educational Use Case:** Locating every mention of technical terminology (e.g., "gradient descent").

Dense Retrieval (Sentence Transformers)

- **Mechanism:** Fixed-dimensional dense vectors via FAISS.
- **Strength:** Semantic similarity and conceptual relationships.
- **Educational Use Case:** Matching concepts without exact wording (e.g., matches "minimizing the loss function by adjusting weights").

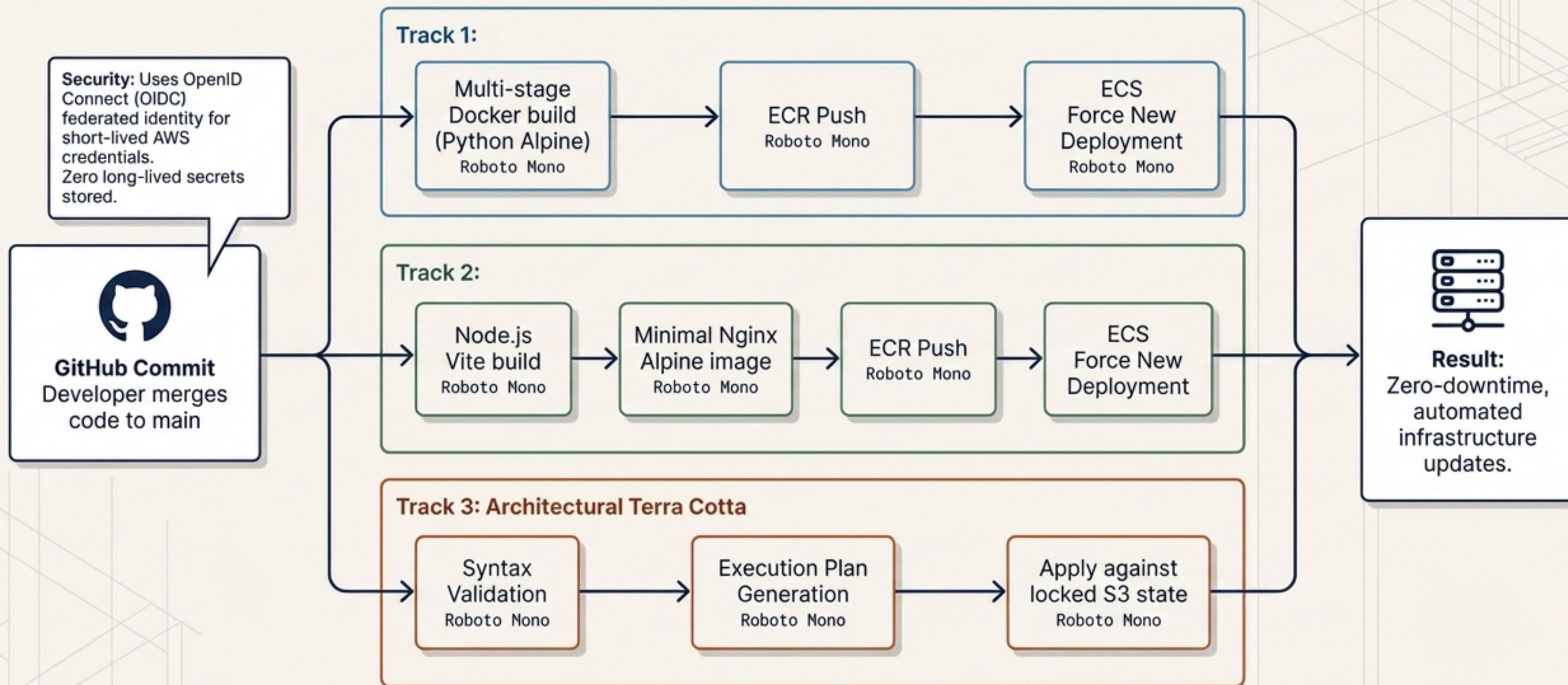
Combined Hybrid Index Outcome

Tunable mixing coefficient merges rankings to capture both precise technical terms and broad concepts for complete discoverability.

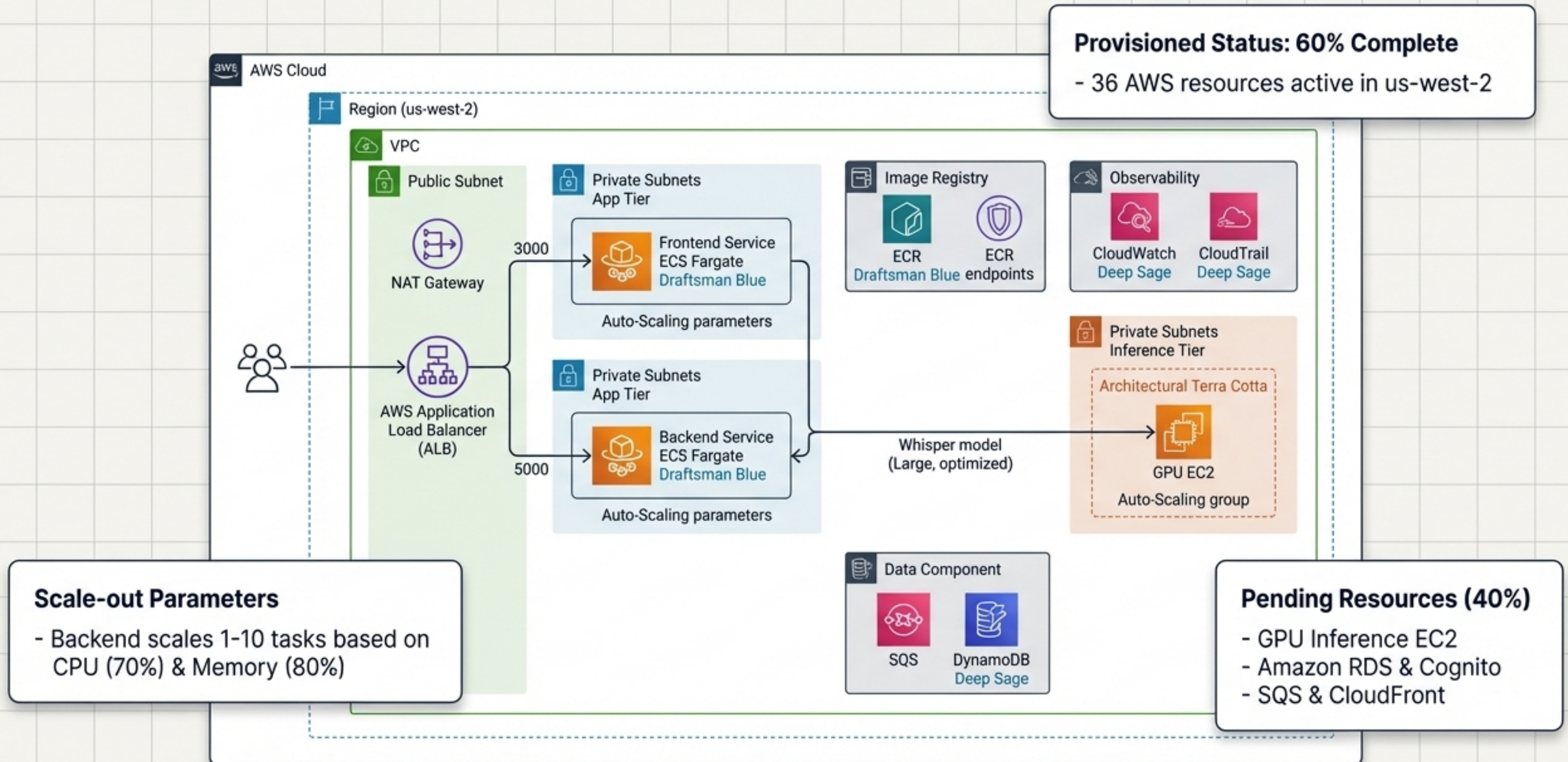
The Excel Stakeholder Matrix

Persona	Typical File Features	Parser Optimization Strategy
Business Analyst (BA)	Multi-sheet, text-heavy matrices, traceability IDs	Prioritize logical tables and cross-references over math.
Data Analyst (DA)	Raw data + PivotTables/Charts, KPI blocks	Preserve worksheet roles, capture dashboard structures.
Project Manager (PM)	Gantt-layouts, date-rich task plans, percent-complete	Handle hierarchical task plans and date normalizations.
Human Resources (HR)	Entity-record tables (one row per employee), status tags	Prioritize stable row-level extraction.
Accountant	Regular financial schemas, subtotals, roll-up formulas	Detect repeated rows and subtotal logic.
CS Student	Ad-hoc benchmarks, log tracking, heterogenous layouts	Robustness to sparse metadata and semi-structured tables.

CI/CD Pipeline to AWS Automation Engine



AWS Infrastructure Mapping (IaC via Terraform)



System Readiness Dashboard

Foundation (NMK)

- ✓ 35 Traced Requirements Documented
- ✓ 10 Interaction flows & Use Cases Mapped
- ✓ Logical System Architecture Finalized

Data/AI Pipeline (NNK & PDZ)

- ✓ 7-Stage Video/Audio Pipeline Operational
- ✓ Zero-Dependency Excel Parser Active
- ✓ Table-Aware Chunker Enabled
- ✓ Hybrid BM25/Dense Indexing Online

Cloud & Deployment (PDZ)

- ✓ Lean Multi-stage Docker Images Built
- ✓ 36 AWS Resources Provisioned via Terraform
- ✓ 3-Pipeline CI/CD Automation Designed

Phase 3 Ready: System is architecturally grounded and implementation-ready.

Next Steps: Phase 3 Trajectory

